THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The next term in 5, 10, 20, 40, … is: 1

(a) 70 **(b)** 100 **(c)** 80 **(d)** 60

2. The 6th triangular number is: 1

(a) 22 **(b)** 19 **(c)** 27 **(d)** 21

3. Which of the following is the smallest angle? 1

(a) 80° **(b)** 100° **(c)** 20° **(d)** 170°

4. An angle of 179° is: 1

(a) acute **(b)** right **(c)** reflex **(d)** obtuse

5. In a winning-strategy game with two players taking 1 or 2 stones, the player who wants to win should aim to leave a multiple of: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 5

6. The Collatz sequence starting at 16 takes how many steps to reach 1? 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 7

7. The most suitable graph for comparing the number of students in different classes is a: 1

(a) rangoli **(b)** calendar **(c)** map **(d)** bar graph

8. In a pictograph where one symbol stands for 12, how many symbols are needed for 72? 1

(a) 7 symbols **(b)** 8 symbols **(c)** 5 symbols **(d)** 6 symbols

9. Which pair is co-prime? 1

(a) 14 and 15 **(b)** 12 and 18 **(c)** 10 and 25 **(d)** 14 and 21

10. The LCM of 3 and 5 is: 1

(a) 30 **(b)** 18 **(c)** 15 **(d)** 10

11. The perimeter of a rectangle with length l and breadth b is: 1

(a) $l \times b$ **(b)** $l + b$ **(c)** $2(l + b)$ **(d)** $4l$

12. Area of a triangle with base 20 cm and height 11 cm is: 1

(a) 130 cm² **(b)** 220 cm² **(c)** 110 cm² **(d)** 31 cm²

13. The simplest form of $\frac{18}{24}$ is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{6}{8}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{9}{12}$

14. The fraction equal to 1 is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{5}{10}$ **(c)** $\frac{5}{5}$ **(d)** $\frac{1}{5}$

15. The point where the diagonals of a rectangle meet is its: 1

(a) vertex **(b)** arm **(c)** centre **(d)** side

16. A square has how many lines joining opposite corners? 1

(a) 3 **(b)** 4 **(c)** 1 **(d)** 2

17. The angle of rotational symmetry of a square is: 1

(a) 45° **(b)** 90° **(c)** 180° **(d)** 60°

18. A scalene triangle has: 1

(a) 3 lines **(b)** 1 line

(c) no line of symmetry **(d)** 2 lines

19. On the number line, a negative integer lies: 1

(a) to the right of zero **(b)** at zero

(c) to the left of zero **(d)** above zero

20. $(-5) + (-3)$ equals: 1

(a) −2 **(b)** 2 **(c)** 8 **(d)** −8

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Write the 12th even number and the 12th odd number. 2

22. What is a supercell in a number grid? 2

23. Name two sources where we can find data in daily life. 2

24. Write all the factors of 17. 2

25. Convert 3 m² into cm². 2

26. Find $\frac{3}{5}$ of 40. 2

27. Name the quadrilateral whose diagonals are equal and perpendicular. 2

28. Write the integer for 20 m below sea level and for a profit of ₹40. 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Explain how the Koch snowflake is built by repeating a rule. 3

30. A wheel turns through 3 right angles. How many degrees is this? Is it acute, obtuse or reflex? 3

31. Explain why we cannot use a scale of 1 unit = 1 student for 2,000 students. 3

32. Why is area measured in square units and perimeter in linear units? 3

33. Why are right angles checked with a set square at the corners of a table? 3

34. A kite has how many lines of symmetry? Explain. 3

35. Show that 143 is composite by finding its factors. 3

OR

Find the smallest 3-digit number divisible by 9. 3

36. The height of a hill is 300 m and the depth of a pit is −50 m. Find the difference. 3

OR

The height of a kite is 40 m above and a fish is 15 m below the ground (water). Find the vertical gap. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) Why do we use degrees for angles? 5
(ii) Why is one full turn 360°?
(iii) Find the angle turned in 1/6 of a turn.

OR

(i) Why are tracks of a railway drawn parallel? 5
(ii) Give two examples of intersecting lines in surroundings.

38. (i) Describe a number game with a winning strategy. 5
(ii) Explain how the first player wins in a game where players take 1 or 2 objects from 10 and the last wins.

OR

(i) Why is the Collatz conjecture famous? 5
(ii) Test it for 13.
(iii) What did you observe?

39. (i) Mention one historical fact about fractions in India. 5
(ii) Write 3/4 in Indian number words.
(iii) Why were fractions needed?

OR

(i) Divide 7 equal cakes among 3 friends. 5
(ii) Write each share as a mixed fraction.
(iii) Explain with a picture.

40. (i) State three regular polygons with their orders of rotational symmetry. 5
(ii) Find the sum of angles of rotation for one full turn in a pentagon.

OR

(i) How are symmetry and reflection related? 5
(ii) Give two examples of rotational symmetry in daily life.

END OF QUESTION PAPER Total: 80 Marks

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